

Exercises Tutorial 9 (23-05-2024)

Exercise 1

It is hoped that a newly developed pain reliever will more quickly produce perceptible reduction in pain to patients after minor surgeries than a standard pain reliever. The standard pain reliever is known to bring relief in an average of 3.5 minutes with standard deviation 2.1 minutes. To test whether the new pain reliever works more quickly than the standard one, 50 patients with minor surgeries were given the new pain reliever and their times to relief were recorded. The experiment yielded a sample mean of 3.1 minutes and a sample standard deviation of 1.5 minutes. Is there sufficient evidence to indicate, at a 5% level of significance, that the newly developed pain reliever does deliver perceptible relief more quickly?

Answer of exercise 1

The natural assumption is that the new drug is no better than the old one. So, the new drug must be proved to be better. Thus if μ denotes the average time until all patients who are given the new drug experience pain relief, the null hypothesis is $H_0 : \mu = 3.5$ and the alternative hypothesis is that $H_1 : \mu < 3.5$. This is tested with significance level $\alpha = 0.05$.

Since the symbol in H_1 is " $<$ " this is a one-sided test and there is a single threshold value, $z_{0.05}$, so that $P(Z < z_{0.05}) = 0.05$. From table A3 we find that $z_{0.05} = -1.645$. So, the critical region is $(-\infty, -1.645]$. The sample is large, but the population standard deviation is unknown (the 2.1 minutes pertains to the old drug, not the new one). Thus we use the test statistic $Z := \frac{\bar{X} - \mu}{S/\sqrt{n}}$, which, by the CLT, approximately has the standard normal distribution. Inserting the data into the formula for the test statistic gives us a realization of $z = \frac{3.1 - 3.5}{1.5/\sqrt{50}} = -1.886$.

The test statistic falls in the critical region, so we reject H_0 . In the context of the problem our conclusion is: The data provide sufficient evidence, at the 5% level of significance, to conclude that the average time until patients experience perceptible relief from pain using the new pain reliever is smaller than the average time for the standard pain reliever.

Remark: We could also have calculated the p-value (in this case it is approximately 0.0294, check this for yourself), which is lower than α , so we reject H_0 .

Exercise 2

In the past the average length of an outgoing telephone call from a business office has been 143 seconds. A manager wishes to check whether that average has decreased after the introduction of policy changes. A sample of 100 telephone calls produced a mean of 133 seconds, with a standard deviation of 35 seconds. Perform the relevant test at the 1% level of significance.

Answer of exercise 2

Notice that the 143 is old population data and it does not come from a random sample. So we want to check if the new population mean is below 143 or not. This gives rise to the following hypotheses: $H_0 : \mu = 143$ vs. $H_1 : \mu < 143$. Level of significance: $\alpha = 0.01$. Test statistic: $Z := \frac{\bar{X} - \mu}{S/\sqrt{n}} \approx \mathcal{N}(0, 1)$ (by CLT). Under $H_0 : Z = \frac{\bar{X} - 143}{35/\sqrt{100}}$. Realization: $z = \frac{133 - 143}{35/\sqrt{100}} = -2.86$. The critical region is $(-\infty, -2.33]$ (table A3). Therefore, we reject the null hypothesis H_0 (P-value: 0.0021).

Exercise 3

The average household size in a certain region several years ago was 3.14 persons. A sociologist wishes to test, at the 5% level of significance, whether it is different now. Perform the test using the information collected by the sociologist: in a random sample of 75 households, the average size was 2.98 persons, with sample standard deviation of 0.82 person.

Answer of exercise 3

$H_0 : \mu = 3.14$ vs. $H_1 : \mu \neq 3.14$ with level of significance $\alpha = 0.05$. Test statistic: $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \approx \mathcal{N}(0, 1)$ (by CLT). Under $H_0 : Z = \frac{\bar{X} - 3.14}{0.82/\sqrt{75}}$. Realization: $z = \frac{2.98 - 3.14}{0.82/\sqrt{75}} = -1.69$. Critical region: $(-\infty, -1.96] \cup [1.96, \infty)$, so fail to reject H_0 (P-value: $2 \cdot 0.0455 = 0.0910$).

Exercise 4

The recommended daily calorie intake for teenage girls is 2,200 calories/day. A nutritionist at a state university believes the average daily caloric intake of girls in that state to be lower. Test that

hypothesis, at the 5% level of significance, against the null hypothesis that the population average is 2200 calories/day using the sample data of size 36, mean 2150, standard deviation 203.

Answer of exercise 4

$H_0 : \mu = 2200$ vs. $H_1 : \mu < 2200$ (one-sided) with level of significance $\alpha = 0.05$. Test statistic: $Z = \frac{\bar{X} - \mu}{s/\sqrt{n}} \approx \mathcal{N}(0, 1)$ (by CLT). Under $H_0 : Z = \frac{\bar{X} - 2200}{203/\sqrt{36}}$. Realization: $z = \frac{2150 - 2200}{203/\sqrt{36}} = -1.478$. Critical region: $(-\infty, -1.645]$, so we do not reject H_0 (P-value: 0.0694).

Exercise 5

Suppose that a delivery company claims that they deliver their packages in 2 days on average, and you suspect it's longer than that. To test this claim, you take a random sample of 10 packages and record their delivery times. You find the sample mean is $\bar{x} = 2.3$ days, and the sample standard deviation is 0.35. Do you have sufficient data to show that the company delivers their packages in more than 2 days, considering significance level 0.05? And what assumptions do you need?

Answer of exercise 5

Because $n = 10$ is much less than 30, and we don't know the population standard deviation, the test statistic must be a t-distribution. This means that we need to assume that the data come from a distribution that is close enough to the normal distribution, so that we can use the test statistic $T = \frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ (recall that a t-distribution is only allowed if the original data is 'approximately normal'). This is in practice an assumption that holds quite often, as long as the distribution is not too extreme and the sample size n is not too small; as discussed in the lecture). The hypotheses: $H_0 : \mu = 2$ and $H_1 : \mu > 2$. Under $H_0 : T = \frac{\bar{X} - 2}{0.35/\sqrt{10}}$. Realization: $t = \frac{2.3 - 2}{0.35/\sqrt{10}} = 2.71$. Critical region: $[1.833, \infty)$, so we reject H_0 .

We can also ballpark the p-value: To do that look into table A4 in the row corresponding to $v = 9$. The test statistic falls between 2 values in this row: 2.574 and 2.821, corresponding respectively to columns 0.015 and 0.01. Thus, we know that the p-value for the test statistic T lies between 0.01 and 0.015. So, we do not know the exact p-value, but because both 0.01 and 0.015 are below $\alpha = 0.05$, we reject H_0 . and we have sufficient evidence that the average delivery time is more than 2 days.

Exercise 6

An old research has determined the mean age of a certain population to be 70.65 with a 95 % confidence interval of (67.2, 74.1). A new sample yields $\bar{x} = 75.32$.

- What is your conclusion concerning the $H_0 : \mu = 70.65$ with $\alpha = 0.05$?
- Is this a one-sided or a two-sided test?

Answer of exercise 6

- Reject H_0 since $\bar{x}$ is outside the confidence interval.
- This is a two-sided test since α is split into two probabilities of 0.025 on either side of the confidence interval.

Exercise 7

The hypothesis $H_0 : \mu = 15$ is tested against $H_1 : \mu \neq 15$ with $\alpha = 0.05$. A random sample has following properties: $n = 20$, $\bar{x} = 17.5$, $s = 5.9$.

- What assumptions should be made and which distribution should be used? Why?
- What is the range of the P-value that you can find with table A4?
- What is your conclusion regarding the null hypothesis?

Answer of exercise 7

- Our distribution must not be too extreme ('approximately normal'). We then use $T = \frac{\bar{X} - 15}{5.9/\sqrt{20}} \sim t_{19}$ (t-distribution with $v = 19$ degrees of freedom) as n is small.

- b) $2P(T = \frac{\bar{X}-15}{5.9/\sqrt{20}} > \frac{17.5-15}{5.9/\sqrt{20}}) = 2P(T > 1.895)$, where $P(T > 1.895)$ is between 0.025 and 0.05. So our P-value is between 0.05 and 0.10.
- c) P-value is bigger than α , thus we do not reject H_0 .

Exercise 8

The average brown trout's I.Q. is 4. Suppose you don't believe this claim. You catch 12 brown trout. A fish psychologist determines the I.Q.s as follows: 5; 4; 7; 3; 6; 4; 5; 3; 6; 3; 8; 5. Conduct a hypothesis test of your belief with $\alpha = 0.05$.

Answer of exercise 8

$H_0 : \mu = 4$, $H_1 : \mu \neq 4$. As our sample isn't large enough to use CLT ($n < 30$), we should use a two-sided t-test with $\bar{x} = \frac{5+4+7+3+6+4+5+3+6+3+8+5}{12} = \frac{59}{12}$, $s = \sqrt{\frac{1}{11} \sum (x_i - \bar{x})^2} = 1.621$. We have to assume that the underlying distribution is not too extreme. The test statistic is $T = \frac{\bar{X}-\mu}{S/\sqrt{12}} \sim t_{11}$. Under H_0 : $T = \frac{\bar{X}-4}{1.621/\sqrt{12}}$. The acceptance region corresponding to a two-tailed t-test with $\alpha = 0.05$ is $(-2.201, 2.201)$. Realization: $T = 1.96$. So we do not reject H_0 .

Exercise 9

A child psychologist says that the average time that working mothers spend talking to their children is 11 minutes per day. Suppose a random sample of 100 working mothers spend on average 11.5 minutes per day talking to their children. The prior research suggests the population standard deviation is 2.3. Do we have evidence that working mothers spend more than 11 minutes per day by talking to their children? Use $\alpha = 0.05$.

Answer of exercise 9

$H_0 : \mu = 11$, $H_A : \mu > 11$. Our test statistic is $\frac{\bar{x}-11}{\frac{2.3}{\sqrt{100}}} = 2.17$, corresponding to the P-value of 0.015, which is less than 0.05, thus we reject H_0 . We have enough evidence to claim that working mothers spend more than 11 minutes per day by talking to their children.

Exercise 10

Consider the previous exercise, but now with a small change: The random sample does not include 100 working mothers, but only 10 working mothers. Will your conclusion change?

Answer of exercise 10

We need to use t-test (as n is small), with $t_9 = \frac{\bar{x}-11}{\frac{2.3}{\sqrt{10}}} = \frac{11.5-11}{\frac{2.3}{\sqrt{10}}} \approx 0.6875$, which in table A.4 (row $v = 9$) lies between values 0.543 and 0.883, which correspond to p-values of 0.3 and 0.2, respectively, which are both bigger than $\alpha = 0.05$. Therefore, we do not have a sufficient evidence to reject H_0 .

Exercise 11

The Achenbach Child Behaviour Checklist is designed so that scores from normal children are Normal with mean $\mu = 50$ and standard deviation $\sigma = 10$. We are given a sample of $n = 5$ children under stress with an average score of $\bar{x} = 56.0$. Is there evidence that children under stress show an abnormal behaviour? You may assume that σ still is known to be equal to 10. Use level of significance 0.05.

Answer of exercise 11

$H_0 : \mu = 50$, $H_1 : \mu \neq 50$ (two-sided), $\alpha = 0.05$. Under the null hypothesis, the stress scores of five children measured $X_1, \dots, X_5 \sim \mathcal{N}(50, 10)$, thus $Z = \frac{\bar{X}-\mu}{\sigma/\sqrt{5}} \sim \mathcal{N}(0, 1)$. Realization: $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{5}} = \frac{56-50}{10/\sqrt{5}} \approx 1.34$, while two-sided test with level of significance 0.05 corresponds to an acceptance region $(-1.96, 1.96)$, as $P(Z > 1.96) = P(Z < -1.96) = 0.025$. Therefore, we do not reject H_0 , i.e. there is not significant evidence of abnormal behavior.

Exercise 12

Solve the previous exercise, assuming that we do not know σ , but are given a sample of $n = 5$ children under stress with an average score of $\bar{x} = 56.0$ and standard deviation $s = 8.5$.

Answer of exercise 12

We have again $H_0 : \mu = 50$, $H_1 : \mu \neq 50$ (two-sided), $\alpha = 0.05$, $\bar{x} = 56$, $s = 8.5$. Under the null hypothesis $T := \frac{\bar{X}-50}{S/\sqrt{5}}$ is distributed according to a t -distribution with 4 degrees of freedom. Realization: $T = \frac{56-50}{8.5/\sqrt{5}} \approx 1.578$ which corresponds in our table A.4 to a p -value between 0.05 and 0.1 for a one-sided t -test, and thus a p -value between 0.1 and 0.2 for the two-sided t -test. This is more than the significance level α , thus we do not reject H_0 . Alternatively: The acceptance region for the two-sided t_4 -distribution and $\alpha = 0.05$ is $(-2.776, 2.776)$ (table A4). Our test statistic clearly lies in this region, so we do not reject H_0 .